Atul Kumar

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Education

Indian Institute of Information Technology Design and Manufacturing (IIITDM) Kurnool **Kurnool, India**
Bachelor of Technology in Artificial Intelligence and Data Science; CGPA: 8.16/10 *Aug 2024 – May 2028 (Expected)*

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Linear Algebra, Probability and Statistics, Object-Oriented Programming, DBMS

Technical Skills

Programming Languages: Python, C++, C, SQL

Machine Learning: Supervised and Unsupervised Learning, Linear and Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, KNN, Naive Bayes, K-Means, DBSCAN, PCA, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Deep Learning: ANN, CNN, RNN, LSTM, GRU, Attention, Self-Attention, Transformers, Transfer Learning, Backpropagation

Computer Vision & NLP: Image Captioning, Feature Extraction, InceptionV3, Sequence Modeling, Tokenization, Word Embeddings

Libraries & Frameworks: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Streamlit

Data & Tools: Exploratory Data Analysis, Data Cleaning, Data Preprocessing, Data Visualization, Git, GitHub, Linux, Jupyter Notebook, Google Colab, Streamlit Cloud

Projects

Image Captioning System using Deep Learning | *Python, TensorFlow, Keras, InceptionV3, LSTM* **Jan 2026 – Mar 2026**

- Developed an image captioning pipeline on the **Flickr8k dataset (8,091 images)**, generating natural language descriptions via transfer learning and sequence modeling.
- Extracted **2,048-dimensional** visual features using a frozen pretrained InceptionV3 encoder and engineered a preprocessing pipeline (90/10 split, minimum word frequency 5, 34-token caption cap).
- Architected a CNN-feature + 256-dim embedding + 256-unit LSTM decoder trained over 20 epochs at batch size 64, evaluating caption quality with BLEU scores. **GitHub:** github.com/atulkumar-ds

Instructor Effectiveness Prediction Model | *Python, Scikit-learn, Pandas, Streamlit* **Nov 2025 – Dec 2025**

- Built a classification model predicting instructor effectiveness from student feedback, achieving **87.5% accuracy** on a **2,000-record** dataset.
- Performed data cleaning and exploratory data analysis on a student feedback survey dataset, engineering features and handling missing values prior to model training.
- Trained and benchmarked **4+** classification algorithms, selecting the final model by cross-validated accuracy and generalization performance.
- Shipped the model as a Streamlit web application serving real-time predictions. **GitHub:** github.com/atulkumar-ds

Startup Funding Analytics Dashboard | *Python, Streamlit, Pandas, NumPy, Matplotlib, Seaborn* **Sep 2025 – Oct 2025**

- Launched an interactive Streamlit dashboard analyzing investment trends and sector-wise funding patterns across **2,821 startup records**.
- Engineered features in Pandas and NumPy and built Matplotlib and Seaborn visualizations of funding distribution and sector-wise investment activity.
- Version-controlled with Git and published to Streamlit Cloud for public access. **GitHub:** github.com/atulkumar-ds/startup-dashboard **Live:** startup-dashboard.streamlit.app

Activities

Smart India Hackathon (SIH) **Kurnool, India**
Team Member *2025*

- Collaborated with a **6-member** team to design and build a functional prototype within a 36-hour development window.
- Demoed the working solution to evaluating judges under fixed time constraints.

College and Online Hackathons **Kurnool, India**
Participant *2025*

- Delivered working prototypes across **3** hackathons under 24–36 hour constraints, dividing implementation work across team members.

Achievements & Certifications

Solved **585+** data structures and algorithms problems across LeetCode, GeeksforGeeks, and CodeChef. Attained **CodeChef 2-Star** (1597, 24 rated contests) and **LeetCode rating 1653**; top 50 at IIITDM Kurnool on GeeksforGeeks.

Coding Profiles: leetcode.com/u/atulkumar_ds | codechef.com/users/atul_35076 | codeforces.com/profile/atulkumar-ds

Certifications: Machine Learning A–Z: AI, Python and R (Udemy) | Data Structures and Algorithms in Python (Udemy)